



Research paper

Hybrid pathfinding optimization for the Lightning Network with Reinforcement Learning

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ABSTRACT

Payment channel networks, such as Bitcoin's Lightning Network, have emerged to address blockchain scalability issues, enabling rapid transactions. Despite their potential, these networks often experience payment failures due to delays in pathfinding, unreliable routes and infrastructure issues, resulting in excessive carbon emissions. Current reinforcement learning solutions for payment channel networks mainly address issues like payment channel balance and routing fees but often overlook the infrastructure-related causes of payment failure. This paper introduces a novel reinforcement learning-based architecture that combines reinforcement learning agent with native deterministic pathfinding algorithms. This hybrid approach leverages the fast, complete solutions of deterministic algorithms while adapting to the network's dynamic and probabilistic nature of payments to significantly enhance payment success rates. Experiments on real network snapshots show that this approach outperforms native pathfinding algorithms and state-of-the-art static optimization methods, providing improved reliability and efficiency in dynamic network conditions. In scenarios with payment failure rates greater than 5%, the proposed approach achieves a 10% higher payment success rate than existing methods, while maintaining balanced performance on economic key metrics such as the payment fee and throughput, and on sustainability key metrics such as payment path length, number of inter-country/continental hops, and average carbon intensity.

1. Introduction

Payment channel networks (PCNs), most notably the Lightning Network (LN), have been widely adopted with the objective of facilitating faster, lower-cost transactions off-chain. Nevertheless, they continue to encounter significant obstacles in attaining consistent success in payment routing. A common criticism of PCNs, and LN in particular, is the inconsistency in payment success rates due to network architecture limitations and a lack of insight into the causes of payment failures (Dasaklis and Malamas, 2023). For example, research conducted by River revealed that payment success rates in their LN infrastructure ranged from 96.5% to 99.6% (River, 2023). However, other reports have indicated much lower success rates, with instances as low as 50% (Stadelmann, 2023). These failures frequently originate from complications associated with pathfinding delays, the absence of optimal routes, node unreliability, and payment channel balancing issues.

A plethora of algorithms have been put forth with the aim of enhancing payment routing and payment path finding in LN. These include deterministic methods such as Flash, Flare, SilentWhispers,

SpeedyMurmurs, ProfitPilot, and Fence (Camilo et al., 2024; Wang et al., 2024) as well as some machine learning architectures (Asgari et al., 2022; Papadis and Tassioulas, 2023; Song et al., 2024). Nevertheless, none have been developed into comprehensive software solutions, nor have they been implemented in LN software clients, due to practical limitations and the necessity for extensive online testing. Furthermore, the efficacy of these solutions has been assessed on relatively modest network samples, which presents a challenge in demonstrating their superiority over native algorithms in real-world scenarios (Chen et al., 2022b). It is notable that the current developments of deterministic algorithms (e.g. Flash) demonstrate inferior performance within a realistic setup in terms of payment success compared to the actual statistics of the LN with its native algorithms (see River (2023)). Consequently, the identification of a more robust and scalable pathfinding solution represents a significant challenge in the pursuit of enhancing the payment success rates in LN.

This paper addresses the aforementioned issues of pathfinding and payment success by employing a novel approach based on reinforcement learning (RL). In contrast to existing RL-based solutions, which

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predominantly address payment success in terms of channel imbalance, fee affordability, or node profitability, this work aims to dynamically optimize pathfinding heuristics in response to network topology and the latent payment failure distribution.

This paper makes the following contributions to the field:

- An extended review of the payment pathfinding solutions and algorithms in the area of PCN is performed, and it is demonstrated that the problem of payment success failures remains understudied.
- A comparison of the native LN pathfinding algorithms is presented, demonstrating that their payment success varies under dynamic conditions and is relatively worse when confronted with a hidden probability distribution of payment failures.
- A reinforcement learning approach is developed for the dynamic optimization of LN payment pathfinding heuristics. This includes the creation of a training environment, a RL-agent prototype, and the design of an efficient reward function.
- The proposed approach is validated using a real network snapshot, and the performance of the trained agent is evaluated in a network where the probability of payment failure is stochastically distributed but dependent on the geographical location of nodes. The proposed approach demonstrates superior performance in terms of payment success while maintaining balance across other key metrics in comparison to the existing methods.

The remainder of this paper is organized as follows. The following section presents an in-depth examination of the current developments in global payment blockchain-based and channel-based networks, identifying research gaps and defining the focus of this work. The *Methods* section presents the methodology that has been adapted for this study, together with a detailed description of the proposed solution, the required data preparation procedures, and the chosen evaluation strategy. The *Results* section presents the findings of the experiments, accompanied by the relevant figures. This is followed by a discussion of the conclusions that can be drawn from the results and suggestions for future work.

2. Literature review and research focus

2.1. Current development of global payment blockchain networks

The advent of global payment networks utilizing blockchain-based cryptocurrency has profoundly transformed the global financial landscape, offering expedited, cost-effective and decentralized alternatives to traditional banking systems. Cryptocurrencies, constructed upon blockchain technology, facilitate peer-to-peer transactions without the involvement of intermediary institutions, thereby enabling cross-border payments with reduced fees and delays. Nevertheless, a number of significant obstacles impede the comprehensive integration of cryptocurrencies into the realm of conventional monetary transactions and the broader financial system. These include concerns pertaining to price volatility, scalability challenges, security vulnerabilities, and environmental impact.

Following the advent of Bitcoin, the inaugural cryptocurrency to exemplify the potential of decentralized payments, in 2009, it was observed that transaction speeds were relatively slow and fees were high during periods of network congestion. This resulted in limitations to its effectiveness as a global payment network. The advent of Ethereum also constituted a notable advancement in the evolution of global payment networks. However, it was still perceived to exhibit inherent scalability constraints, vulnerabilities pertaining to smart contracts, and centralization concerns, as evidenced by the findings of [Mehtar et al. \(2019\)](#).

In order to address these challenges, developers and researchers have pursued two principal avenues of inquiry. The first of these involves the proposal of technological enhancements to the existing

infrastructure and algorithms, alongside the development of the cryptocurrency ecosystem through the introduction of novel computation methods. The second entails the introduction of new architectural solutions, which may encompass a range of approaches, from incremental improvements to consensus algorithms to the establishment of novel cryptocurrency networks and the development of innovative network designs, as well as the exploration of new architectural programming patterns.

Despite the extensive adoption of cryptocurrencies, their value remains highly volatile, which has a significant impact on investment and trading decisions. This has led to extensive research into the prediction of their price. In the first direction of development, recent papers have presented improved variants of the sine-cosine algorithm and meta-heuristics for accurate Bitcoin price prediction (e.g., [Salb et al., 2022](#); [Mizdrakovic et al., 2024](#)). These include the use of recent advances in neural networks (e.g., [Milicevic et al., 2023](#); [Nayak et al., 2022](#)). In the second direction, price volatility has been managed primarily through the introduction of projects such as Stablecoin, including Tether (USDT) and USD Coin (USDC), which have gained traction as payment solutions due to their higher price stability.

In view of the limited efficiency of the existing cryptocurrency infrastructure, which is affected by network congestion and low transaction speeds, a number of new architectural solutions have been proposed. The most significant of these are changes to consensus algorithms, which represent a core technology of the blockchain. A number of highly effective consensus algorithms, including Proof of Stake, Delegated Proof of Stake and others, have been proposed (for a review, see [Xiong et al., 2022](#); [Hussein et al., 2023](#)). The selection of a consensus algorithm entails a number of trade-offs, with the choice of an algorithm potentially having significant implications for the performance and stability of a blockchain network. Furthermore, given that the architectural characteristics of a blockchain inform significant design decisions, a number of architectural patterns have been identified for blockchain application software architectures. These are intended to address the unique challenges presented by blockchain technology. For further details, please see, for example, [Alzhrani et al. \(2023\)](#) and [Xu et al. \(2017\)](#).

A contrasting perspective is offered by the second direction of development, which concerns the design of “layer-two” networks constructed on top of existing (layer-one) cryptocurrency infrastructures. Examples of such infrastructures include the Lightning Network ([Poon and Dryja, 2016](#)) for Bitcoin and the Raiden Network ([Raiden, 2024](#)) for Ethereum. In contrast with the aforementioned solutions, layer-two protocols facilitate the scaling of blockchains without modifying the underlying layer-one trust assumptions and do not extend or replace the existing consensus mechanisms ([Gudgeon et al., 2020](#)).

Layer-two protocols facilitate the execution of transactions, designated as *off-chain* through the utilization of private and authenticated communication channels, as opposed to broadcasting each transaction on the parent blockchain. The advent of layer-two networks and a novel category of blockchain-related payment networks, designated as PCNs, has resulted in a notable decline in the transaction load on the underlying blockchains. However, there is potential for further enhancements in aspects such as privacy and network performance optimization, as evidenced by [Dasaklis and Malamas \(2023\)](#).

2.2. Current development of global payment channel networks

The fundamental premise of PCNs and LN, as a case in point, is to facilitate the majority of payment transactions outside the blockchain, thereby enabling nodes to establish two-way (bidirectional) payment channels ([Wikipedia, 2024](#)). Each payment channel functions as a joint account for the respective channel members. To guarantee the security of the system, the opening of a payment channel necessitates the transfer of funds to a shared address on the blockchain. Such funds serve as a secure repository for potential multiple off-chain transactions

between the two parties. Upon the closure of a payment channel, the final balance is settled on the underlying blockchain (Avarikioti et al., 2023).

This approach circumvents the dissemination of each blockchain transaction to the entire network. Conversely, they employ the costly and low-yield blockchain solely as a means of resolving disputes. PCNs promise to conduct off-chain transactions in seconds, rather than minutes or hours, while maintaining the security of assets, reducing fees and improving the scalability of the blockchain (Gudgeon et al., 2020).

It is also noteworthy that each node in such a network is capable of establishing payment channels with numerous other nodes. This results in the formation of a network that enables the transfer of funds to neighboring entities via the intermediary of other peers. The establishment and maintenance of a payment channel necessitates the blocking of funds, thereby generating opportunity costs for the intermediary. In order to reduce these costs, intermediary nodes receive a commission (fee) for their services.

The mechanics underlying PCNs have attracted considerable attention in recent literature (for a review, see Gudgeon et al., 2020). Furthermore, the cryptographic foundations of PCN security proofs have been subjected to analysis (Malavolta et al., 2017; Beres et al., 2020; Kappos et al., 2021). Additionally, efforts have been made to comprehend the game-theoretic aspects of these networks from two distinct perspectives: security (Hüttel and Staroveski, 2020; Avarikioti et al., 2020) and economic (Avarikioti et al., 2023; Van Engelshoven and Roos, 2021).

In addition to the LN, which will be discussed in detail in the following section, a number of PCNs have been developed as continuations of the LN concept for other blockchain networks (for a detailed review, see Khojasteh and Tabatabaei, 2021). The most notable of these are:

- The Raiden Network (Raiden, 2024) adapts the LN concept for Ethereum, thereby reducing transaction costs by up to seven times compared to native Ethereum. Raiden is an efficient system for processing micropayments and supports off-chain transactions with any token that adheres to the standard Ethereum token API.
- The Sprites Network (Miller et al., 2019) is an Ethereum-focused network inspired by the LN and Raiden, but with the objective of reducing collateral costs in off-chain payments. In contrast to Lightning and Raiden, where the cost of collateral increases in line with the length of the payment chain, Sprites maintains a constant cost by leveraging Ethereum's smart contracts.
- Ripple (Schwartz et al., 2014) is a payment settlement system and cryptocurrency exchange network that is capable of processing transactions on a global scale. It does not employ blockchain technology to achieve consensus, instead it utilizes a hash tree to summarize data and compare this across its validating servers.
- Duplex Micropayment Channels (Decker and Wattenhofer, 2015) employs Bitcoin's timelock functionality (Roche, 2019) to facilitate routed payments with minimal delay.
- Teechan (Lind et al., 2019) is a full-duplex payment channel framework on Bitcoin for micropayments. The system employs multi-signature time-locked transactions, thereby enabling concurrent payments in both directions.
- Plasma (Poon, 2017) is a framework for the execution of smart contracts on the Ethereum platform, utilizing a hierarchical structure of blockchains. Child chains are secured by parent chains, and computation is framed using *MapReduce* functions.

There is sufficient empirical evidence to suggest that the most common PCNs, such as LN, Reiden, and Ripple, exhibit similarities in their distribution, topology, and dynamics (Rohrer et al., 2019; Seres et al., 2020; Rebello et al., 2022; Chen et al., 2022a). Consequently, these networks can be approximated by *scale-free* networks (Seres et al., 2020).

As evidenced by the relevant literature, the LN is the most impactful PCN, with numerous current and potential business-related

applications. These include micropayments, cross-border and remittance payments, IoT and machine-to-machine payments, decentralized exchanges, transfers of digital assets and LN-related services (Dasaklis and Malamas, 2023). As of 28 May 2024, the LN comprises 51,769 observed nodes and 51,925 payment channels (see Fig. 1), with an estimated network capacity of 4851.75 BTC (Stat1ml, 2024). The considerable success of the LN has guaranteed its position as the foremost PCN implementation (Camilo et al., 2022).

2.2.1. Lightning network topology and dynamics

The preeminent position of the LN necessitates a comprehensive examination of its growth and topological characteristics (see, for example, Camilo et al., 2022; Valko and Kudenko, 2024; Seres et al., 2020), including the formulation of novel topological improvements (see, for example, Mahdizadeh et al., 2023). Furthermore, numerous studies have also identified shortcomings in the current structure of the network, particularly in terms of its vulnerability to various attacks, including those of a topological nature (see, for example, Rohrer et al., 2019). It is therefore crucial to gain an understanding of the topology and dynamics of the network in order to ascertain the viability of optimization based on the identified patterns and related heuristics.

The LN is an evolving network, its growth being contingent upon the addition of new users to the network and the establishment of payment channels with existing nodes. The considerable scale of this network is illustrated in the figure below (see Fig. 1).

In the most recent analytical paper (Mahdizadeh et al., 2023), the average degree of nodes in the LN was found to be 8.46, with a high degree of skewness, and the density was determined to be 0.00046. This indicates that the network is relatively sparse. The effective diameter is approximately 12, and the average shortest path length in the network is 3.65. Although this calculated value is independent of the payment channel capacities, it indicates that nodes are highly accessible and that multi-hop payments with excessive hops are absent in the network (Mahdizadeh et al., 2023).

The number of bridges is approximately 8560 payment channels. As a consequence of the existence of a considerable number of nodes with a degree of 1, a significant proportion of the calculated bridges are edges associated with single-degree nodes. The number of bridges in the network, excluding those associated with single-degree nodes, is 174. This suggests the existence of a substantial number of crucial payment channels for maintaining network connectivity (Mahdizadeh et al., 2023).

The local clustering coefficient of a node is an indicator of the level of connectivity among its neighboring nodes. This value indicates the degree of similarity between the sub-graph formed by a node's neighbors and a clique comprising those neighboring nodes. The mean value of local clustering coefficients for network nodes is 0.1231. This value, excluding zero values, would be 0.3255 (Mahdizadeh et al., 2023). The assortativity coefficient of degrees in a network indicates the tendency of nodes to form connections with other nodes that have similar degrees. This characteristic is measured on a scale between 1 and -1 , where a positive value indicates a tendency for high-degree nodes to connect with other high-degree nodes, and a negative value indicates a tendency for high-degree nodes to connect with low-degree nodes. In the LN, the assortativity coefficient of degrees is -0.195 , as indicated in Mahdizadeh et al. (2023). This value reflects the interconnection of low-degree nodes with high-degree nodes.

In accordance with the principles of network dynamics, the LN is observed to incorporate, on average, 4%–5% of novel nodes and 10%–15% of novel payment channels on a monthly basis. It is notable that the majority of these channels serve to supplant existing ones (Valko and Kudenko, 2023). It was found that almost 50% of nodes could be theoretically reached by two payment hops, 93% by three hops, and almost 98% by no more than five hops (Shell, 2022; Beres et al., 2020). This also represents an advantage in terms of dynamic stability, which can be utilized by intelligent agents or other approximation algorithms

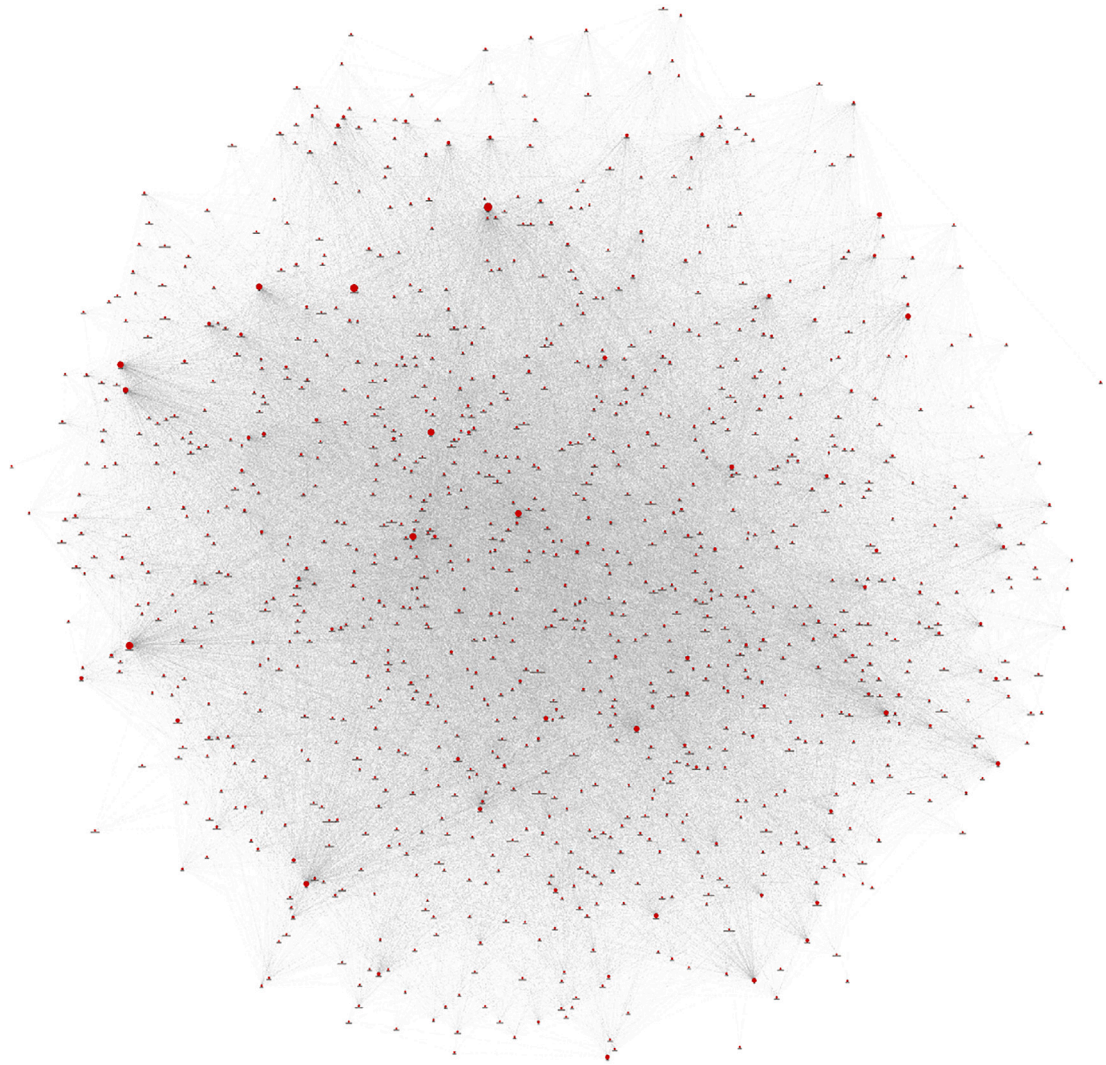


Fig. 1. Lightning Network graph visualization.

Note. The Lightning Network graph visualization of nodes and payment channels, as of 23 May 2024 (LNRouter, 2024). For the sake of clarity, only nodes with more than 15 payment channels are shown.

for performance optimization purposes.

It can thus be concluded from these topological characteristics that the LN exhibits certain patterns that can be approximated using admissible heuristics to perform some payment pathfinding algorithms with greater efficiency. This conclusion is consistent with the results of other recent studies (Valko and Kudenko, 2024).

2.2.2. Lightning network payment success challenges and the focus of this work

A common criticism of PCNs and, in particular, the LN is that payments do not always succeed (Dasaklis and Malamas, 2023). As documented in River's research report (River, 2023), the average payment success rate on their LN infrastructure ranged from 96.5% to 99.6% (as a maximum in 2023). Other sources have indicated a success rate of approximately 50% (Stadelmann, 2023). This is a challenging problem to fully resolve, particularly when developers have limited insight into the underlying causes of payment failures due to the intrinsic architecture of the network.

Generally speaking, payment failures in PCN can occur due to several factors, the most significant are:

- **Insufficient funds:** A common cause of payment failure is when the sender lacks adequate outbound capacity to cover the payment amount. In PCNs, only the sender's outbound balance is spendable (Waugh and Holz, 2020; Pickhardt and Richter, 2021).
- **Inadequate payment channel connectivity:** If the sender's payment channels do not connect to the recipient, the payment cannot be routed successfully (D'Angelo et al., 2016; Pickhardt and Richter, 2021; Davis and Harrison, 2022). Maintaining connections with well-established peers – characterized by high uptime, substantial capacity, and numerous channels – can enhance the likelihood of successful payments.
- **Large payment sizes:** Larger payments are more prone to failure due to the increased difficulty in finding a single payment path with sufficient liquidity (Waugh and Holz, 2020; Pickhardt and Richter, 2021).
- **Channel depletion:** Over time, payment channels may become unbalanced or depleted, especially if used predominantly in one direction. This imbalance can lead to payment failures (Podiatchev et al., 2024; Rohrer et al., 2019; Pickhardt and Richter, 2021).
- **Network attacks:** Malicious actors can intentionally cause payment failures through various attacks. Implementing robust security measures and staying informed about potential vulnerabilities

are crucial for network integrity (Khalil et al., 2023; Weintraub et al., 2021; Rohrer et al., 2019; Kappos et al., 2021).

- Pathfinding and routing¹ algorithm limitations: The efficiency of the subsequent pathfinding and routing algorithms plays a significant role in payment success (Weintraub et al., 2021; Pickhardt and Richter, 2021). Employing more resilient algorithms and routing protocols can mitigate various circumstances such as a lack of suitable routes, node unreliability, balancing issues etc.

Current software implementations of the LN protocol have mainly focused on three strategies for handling mentioned uncertainty of sending a payment (Pickhardt and Richter, 2021): (i) incentivizing the pathfinding algorithm to favor larger channels, (ii) ad-hoc splitting of large payment amounts into smaller ones after failed attempts using a technique called multi-part payments, (iii) using provenance scores of nodes and channels and other data collected during operation of a node to estimate which nodes and channels might be reliable.

These strategies are not a panacea, because of naive design choices in three main areas (Khojasteh and Tabatabaei, 2021; Sivaraman et al., 2020): (i) payment pathfinding and routing, (ii) timing of payment transaction submission and its final success, (iii) network deadlocks. Firstly, PCNs such as the Lightning and Raiden Network utilize *source-routing*, whereby clients typically select the shortest path from source to destination. This approach has the potential to degrade throughput for two reasons. First, shortest-path routing can result in congestion in some payment channels, while others remain underutilized. Second, the dominance of funds flowing in one direction can lead to imbalanced payment channels, with the majority of funds concentrated on one side (Sivaraman et al., 2020).

Secondly, the majority of PCNs operate with circuit switching, whereby transactions are processed immediately and independently upon arrival. This results in a number of issues, including exceeding the available balance on the selected payment path and imbalance problems for large transactions (Khojasteh and Tabatabaei, 2021).

Thirdly, in certain instances, concurrent transactions may interfere with one another, resulting in the network becoming unresponsive despite the availability of sufficient endpoint balance.

Addressing these issues involves ensuring sufficient liquidity, establishing connections with reliable peers, utilizing payment-splitting techniques for large transactions, regular or pro-active rebalancing channels, implementing strong security practices, and adopting efficient routing algorithms. So far, none of the existing solutions have been practically implemented to be an improvement over the LN native algorithms.

In summary, PCNs are designed to facilitate high transaction throughput. It is crucial for routers (intermediate nodes) to economically balance the opportunity costs in payment channels and encourage end-users to accept transactions by ensuring an adequate quality of payment services. A transaction will only be successful if all payment channels along its path have sufficient funds and all nodes involved in the transaction are accessible in terms of network connectivity and transport layer availability. It is therefore evident that routing protocols and payment pathfinding algorithms, which are employed to identify an optimal path for a payment transaction, are of paramount importance.

The current pathfinding methods employed in the LN for routing payments are discussed in Kumble and Roos (2021), namely LND (LND,

2024), c-Lightning (CLN) (CLN, 2024) and Eclair (ECL) (ECL, 2024). They are Dijkstra-based and *native* for LN client software. Such algorithms utilizing source-routing have also been the subject of criticism due to their limited scalability in large networks (Grunspan and Pérez-Marco, 2019). A number of deterministic algorithms have recently been developed with the aim of addressing these issues, including: Flare (Prihodko et al., 2016), SilentWhispers (Malavolta et al., 2016), SpeedyMurmurs (Roos et al., 2017), Flash (Wang et al., 2019), ProfitPilot (Camilo et al., 2024) and Fence (Wang et al., 2024). It is regrettable that neither of these solutions has been developed into a complete software package, nor implemented in LN software clients. This is due to the necessity for extended online testing and the potential need for topological or infrastructural changes, which may prove unfeasible. Furthermore, the most prominent of these have been evaluated on relatively small samples of a network (e.g. 128 (Wang et al., 2024), 512 (Camilo et al., 2024), 1870 (Wang et al., 2019), 1900 (Chen et al., 2022b) nodes). This is insufficient to prove superiority over native algorithms in the real world. Furthermore, it has been asserted that none of the devised algorithms in PCNs are capable of meeting the desired criteria of scalability, efficiency and effectiveness under real dynamic conditions (Gudgeon et al., 2020; Chen et al., 2022b).

It is therefore of great importance to address the aforementioned issues and to achieve a higher level of payment success in the current development of PCNs, particularly in the context of the LN. It is of particular importance to pursue this objective through the utilization of architecturally less expensive methodologies, such as the modification of pathfinding heuristics (Valko and Kudenko, 2024). In the case of the LN, it is crucial to develop heuristics that are robust and take into account the network topology, as well as the uneven utilization of its infrastructure, during the pathfinding phase (Valko and Kudenko, 2024). It is therefore the objective of this work to conduct further research into the development of effective payment pathfinding heuristics, with a view to improving the payment success rates.

Given the inherently dynamic nature of payment success in LN infrastructure, with success rates depending not on financial metrics alone but also on technical characteristics such as transport layer network performance and node and payment path reliability (River, 2023), this paper presents and evaluates a novel approach to dynamically optimize pathfinding heuristics using reinforcement learning.

2.3. Reinforcement learning for the pathfinding task in payment channel networks

The application of the RL-based approach to address pathfinding tasks in diverse network configurations has been extensively explored in the literature (for a comprehensive review, see Mammeri, 2019). For example, Boyan and Littman (1993) pioneered the integration of the Q-learning algorithm with packet routing, enabling the dynamic learning of routing scenarios and the identification of the shortest path within a network. Gao et al. (2017) put forth a multi-agent routing algorithm that employs Q-learning and backpressure. This approach requires each routing node to possess only local information about its neighboring nodes, thereby enabling the effective resolution of the aforementioned task. Mayadunna et al. (2017) and Yang et al. (2019) put forth a RL-based approach to detect malicious routing nodes in mobile ad hoc networks and wireless sensor networks, respectively. Nevertheless, these developments have been largely confined to specific network types such as wired networks, wireless networks, ad hoc networks etc., and certain sub-types of peer-to-peer networks like IoT blockchain-based networks, and have only recently begun to be applied to PCNs. This specificity means that the definitions of state and action spaces, as well as the design of features, reward functions, and even software implementations, are typically tailored to the unique topological and even physical characteristics of a certain type of network. This implies that these RL-based solutions cannot be considered universally applicable or proven across different network categories.

¹ The pathfinding phase involves identifying an optimal route for a payment transaction based on network topology and constraints like fees and liquidity. It is a computational process that occurs before payment execution. In contrast, the routing phase is the real-time execution of the payment along the chosen route, ensuring funds are transferred while adapting to dynamic network conditions. In this work, we focus on the pathfinding solution; however, our evaluation also simulates the routing phase, as these two phases are dependent and influence the final payment success.

In recent years, numerous attempts have been made to address the dynamics of PCNs using a range of RL-based techniques and solutions (see Table 1). For instance, Kadry and Gadallah (2021) introduced a modified RL-algorithm focusing on payment path selection, achieving improved routing success by excluding inactive nodes and updating Q-values for paths as nodes are removed. Luo and Li (2022) used multi-agent deep Q-learning approach (DQN) to prioritize transactions based on historical success and fee data, enhancing network throughput, although pathfinding task was handled by existing algorithms. Chen et al. (2022a) developed PLAC, an actor-critic model optimizing throughput by managing router node transactions, yielding a throughput increase of up to 34.9% and outperforming Flash algorithm. Davis and Harrison (2022) proposed an RL-model for improving node connectivity in small PCN networks. Chen et al. (2024) proposed DRL-PCR, a deep RL approach to rebalance payment channel capacity, achieving about 40% payment success with the average LN configuration but without addressing the network dynamics. Many recent studies have also aimed to increase the profitability of the node or security of payments without addressing the pathfinding task itself (e.g., Asgari et al., 2022; Papadis and Tassioulas, 2023; Song et al., 2024; Qiao et al., 2024).

The most proximate to the present work is the recent introduction of a method designated as one-step path prediction by Valko and Kudenko (2023). They presented an RL-based solution for a network agent that learns its neighborhood and utilizes local knowledge to address the payment pathfinding problem. Although this preliminary solution outperforms native algorithms in terms of execution time by reducing computational cost at each pathfinding step, it was designed and tested on a relatively small sample of the network snapshot (up to 100 nodes). Furthermore, the payment success metric, defined as the number of transactions successfully completed by an agent within its own neighborhood, could only be evaluated to assess the agent's capacity to adapt to changes in network topology. In the absence of substantial changes in the neighborhood, the agent achieves a payment success rate of approximately 96%. However, the solution lacks scalability and cannot be deployed in a real-size network where transactions are conducted sequentially across the entire graph. Consequently, a fair comparison cannot be made with the current work.

It can be concluded that the infrastructural roots of payment failures remain uncovered, given that existing RL-based solutions mainly consider the payment success as a result of payment channel imbalance or fee affordability and routing profitability. Furthermore, none of the aforementioned works has considered the payment pathfinding problem itself (with the exception of Valko and Kudenko, 2023) or how to obtain candidate payment paths in an effective manner. There is still considerable scope for substantial optimization (Chen et al., 2022a). It should be noted that only work (Valko and Kudenko, 2023) conducted a comparison of the proposed solution with the native pathfinding algorithms (LND, CLN and ECL) that are currently in use on a full-scale basis.

It is important to note that most RL-based solutions (see Table 1) can be classified as *integrated*, meaning the RL-agent operates downstream and makes high-level decisions (such as payment path selection, fee setting, or hops prioritization) based on the output of a deterministic pathfinding algorithm. Only two studies propose *independent* solutions, where the RL-agent makes decisions based solely on the state of the network, without relying on deterministic algorithms in its pipeline. This independent approach can be more efficient in terms of computational complexity and runtime during the payment pathfinding phase. However, due to the large and complex nature of real-world PCNs, this approach often faces scalability challenges, such as computationally expensive state representations and a vast action space.

The approach proposed here is instead a *hybrid* architecture, where the RL-agent operates upstream, directly influencing the heuristic used by a deterministic algorithm during the pathfinding phase. This hybrid approach avoids the scalability issues inherent in fully independent approaches and enables dynamic optimization across various stochastic

metrics. However, it remains slightly less efficient in terms of pathfinding execution time, as deterministic algorithms are still part of the pipeline.

In contrast to the mentioned works, this paper seeks to address the issue of payment success by considering the underlying distribution of payment failures within a network and dynamic balancing across the set of sustainability metrics. This is done by performing a comprehensive comparison with native LN pathfinding algorithms using a real-world snapshot of the LN.

3. Methods

3.1. Reinforcement learning preliminaries and the design of the proposed solution

The RL-based algorithm considers a general situation in which an agent acts as a decision maker and seeks to make decisions in an interactive environment. At each decision point, the environment transmits the current state to the agent. In response, the agent executes an action in accordance with its policy. Subsequently, the environment returns a reward, which serves as an evaluation of the efficacy of the action. Furthermore, the environment progresses to a new state in accordance with the action executed. Such a setting should satisfy the requirements of a Markov decision process (MDP). A MDP is defined as a tuple, $M = (S, A, \mathcal{T}, R, \gamma)$, where S and A are the state and action space, respectively. The transition function, denoted by $\mathcal{T}(s, a, s') : S \times A \times S \rightarrow [0, 1]$, is a probability distribution over the set of states, which describes the likelihood of reaching a state s' from a given state s after executing an action a . The reward function, denoted by $R(s, a, s') : S \times A \times S \rightarrow R$, assigns a numerical reward to a state transition from s to s' with respect to the executed action a . A policy, defined as a function $\pi(s, a) : S \times A \rightarrow [0, 1]$, specifies the manner in which the agent should act within the environment, through the establishment of a probability distribution over all actions in every state. The decay factor, defined as $\gamma \leq 1$, is employed to ascertain the anticipated discounted return and the value function, designated as $v(s)$. In light of the aforementioned, the agent is tasked with learning the policy $\pi(s, a)$ that maximizes the expected discounted return. It is established that the optimal policy is that which exhibits the maximum expected discounted return (Sutton and Barto, 2018).

An off-chain PCN, in particular the LN, can be modeled as a graph, denoted by $G = (V, E)$, where V represents the set of vertices (i.e. nodes) and E represents the set of edges (i.e. payment channels). To complete a payment demand, the sender initiates the set (one or more) of PCN transactions, denoted by T . For each transaction $t \in T$, the sender is responsible for determining the candidate payment paths, denoted by P , to send those transactions. Each candidate path, represented by the sequence of routing nodes $p = [u, \dots, v]$, is defined as a path from the sender node u to the receiver node v , with the condition that $u \neq v$. It is the responsibility of each routing node to select a candidate payment channel, or alternatively a sequence of payment channels, to forward the payment. For simplicity, the predicted payment path p can be determined by selecting the cheapest channels among a sequence of routing nodes $[u, \dots, v]$.

In the proposed solution, the PCN-graph G is treated as the core component of the environment, which implements the agent as a mapping from states to actions to be taken. It is evident that at each stage of the process, the agent receives a transaction t , together with the associated source and destination nodes u, v . The agent then requests a state, designated as $s_{u,v} \in S$, which is constructed by the environment from a so-called neighborhood observation (Fig. 2). The state is defined by a vector subset of the features of the nodes adjacent to u and v , with a total of n features. From a technical standpoint, the value of n can be less than or equal to the number of existing nearest neighbors of u and v , multiplied by the number of their observed features.

Table 1
RL-based techniques and solutions for LN.

Solution	Task	Goal	Validation network size	Tested network model/ scenario	Solution architecture	Reported average success rates in best/average setup	Sustainability considerations	Comparison with native algorithms
MARL (Kadry and Gadallah, 2021)	Payment path selection. The set of potential payment paths is determined by Breadth-First Search.	Select a payment path with the lowest transaction fees.	Simulated full-sized network of 80,591 nodes and 245,394 edges.	The effect of increasing the number of offline nodes and payment channel imbalances.	Integrated	95/80% MARL	–	–
DRL-PCR (Chen et al., 2024)	Payment channel rebalancing request prediction. The set of candidate paths is defined by the Dijkstra shortest path or Flash algorithm.	Rebalance payment channels to improve transaction throughput and success ratios.	Full-sized network snapshot of 9021 nodes and 128,984 payment channels.	The effect of scaled channel balances and the growing amount of transactions.	Integrated	82/40% DRL-PCR+Flash	–	–
LPS (Luo and Li, 2022)	Priority assignment problem. The candidate payment path is defined by LND.	Prioritize hops over the payment path to maximize the transaction rate and minimize the forwarding fees.	Small-world sample of 18 nodes and 36 payment channels.	The effect of manipulated forwarding payment channel capacity.	Integrated	92/78% LPS+LND	–	+/-
PLAC (Chen et al., 2022a)	Defining the maximum amounts to be sent through the payment channels between router nodes.	Maximize long-term throughput.	Full-sized network snapshot, but the size is not reported. 40 nodes under PLAC control.	The effect of scaled payment channel balances and the number of nodes under control.	Integrated	47/7% ^a PLAC+EDJ	–	–
ReBEL (Papadis and Tassiulas, 2023)	Developing payment channel rebalancing policy.	Maximize the node long-term average expected profit from fees.	Simulated network, the size is not reported.	PCN-network simulator, varying fee proportion.	Integrated	–	–	–
BiLSTM-PPO (Song et al., 2024)	Fee-setting problem.	Achieve the optimal fee settings and revenue for nodes.	Simulated network of 100-400 nodes.	Random transactions around one simulated node.	Integrated	–	–	–
RLPath (Qiao et al., 2024)	Payment channel balance prediction, subsequent pathfinding.	Accurately discover all payment channel balances.	Network snapshots of 513–3732 nodes and 8474–55,428 balances.	The effects of graph partitioning methods and parallelization.	Independent	–	–	–
DyFEn (Asgari et al., 2022)	Maximize income for the node.	Define possible fees for payment channels of the node.	Simulated network of 100–500 nodes.	Simulated transactions across 3 random nodes.	Integrated	–	–	–
MBI-A2C (Davis and Harrison, 2022)	Connection point recommendation.	Propose nodes to open payment channels with, to maximize the node expected routing opportunities.	Simulated network of 128 nodes.	Scenarios with different openings budget.	Integrated	–	–	–

(continued on next page)

Table 1 (continued).

RLA (Valko and Kudenko, 2023)	Payment pathfinding.	Find an acceptable payment path with less runtime.	Sampled sub-networks of 50–100 nodes.	Random transactions in sub-networks of different sizes.	Independent	96/78%	+	+
Current work	Payment pathfinding.	Find payment paths that are expected to be successful with balanced sustainability metrics.	Full-sized network snapshot of 9890 channels and 5966 nodes.	Varied the average probability of payment channel failure within a range of 1% to 6%.	Hybrid	100/66%	+	+

^a Normalized throughput instead of payment success rates is reported.

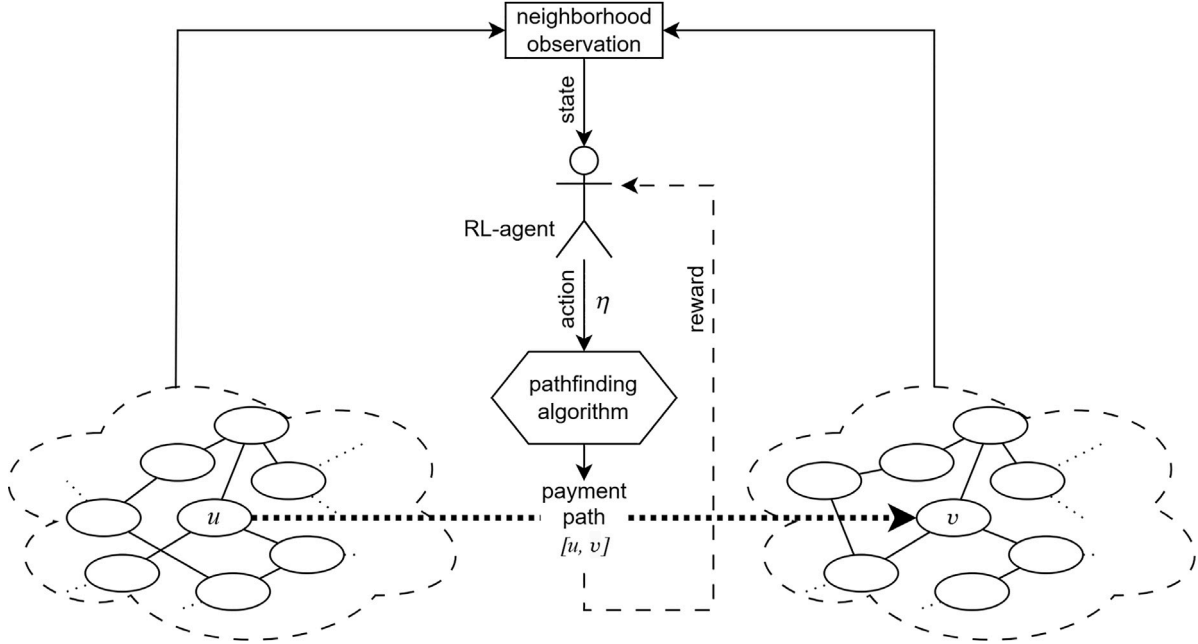


Fig. 2. The proposed agent-based design of the network pathfinding engine.

In principle, based on the observed features of neighboring nodes and payment channels for a given transaction t , as well as the information about subsequent successful payments, the agent is able to discern the hidden dynamic and topological characteristics of the given neighborhood sub-networks. This local knowledge will permit the agent to further modify the pathfinding heuristics that have been previously employed in order to dynamically predict and improve payment success. Accordingly, in the proposed implementation, the agent's selected action, represented by $a_{u,v} \in A$, returns the value of the modifier, denoted by η , $a_{u,v} = \eta(u, v)$. This value will subsequently modify the pathfinding heuristic, represented by h , at the next payment pathfinding step (see Fig. 2).

As shown in Fig. 2, this paper presents a hybrid architecture based on reinforcement learning that combines a trained agent and a native deterministic pathfinding algorithm. This novel hybrid approach utilizes the fast and complete pathfinding solutions provided by Dijkstra-based deterministic algorithms, while adapting to the dynamic and probabilistic nature of payments in the network.

In light of the aforementioned definitions and the availability of both observable payment channel characteristics (payment delay, fee ratio) and sustainability-related payment path metrics (estimated geographical distance, carbon intensity, number of inter-country and inter-continental hops) as an implemented state representation $s_{u,v}$, it can be posited that a well-trained agent should be capable of outperforming native static heuristics in such a dynamic environment.

Given that, the current state representation $s_{u,v}$ is a matrix containing the characteristics of payment channels, derived from up to k available payment channels established between up to m neighboring nodes of source u and destination v nodes, see Fig. 3.

To determine the set of neighboring nodes, we use standard ego-graph sampling² with a radius of m . Given that the effective LN diameter does not exceed 8 hops, on average, and after testing different k -values to achieve reasonable learning speed, we found an efficient configuration with $k = 15$ and $m = 5$. This combination allows the agent to develop admissible pathfinding heuristics using limited and dynamic nearest neighbors information, which is a common situation in real-world payment networks.

In order to maintain consistency with the preliminary results obtained earlier using static multi-criteria optimization (for further details, please refer to Valko and Kudenko, 2024), the reward function r is defined as the weighted payment success of the chosen action, as outlined in Eq. (1).

$$r(a_{u,v}) = \frac{\text{success}([u, v]) \cdot \text{dist}([u, v]) \cdot 10^3}{\sum \text{int}_{\text{CO}_2}[u, v]}, \quad (1)$$

where $\text{success}([u, v])$ is equal to 1 if the payment was successful, 0 otherwise; $\text{dist}([u, v]) = |p_{u,v}|$ is the payment path length; int_{CO_2} is the

² Ego-graph sampling returns induced sub-graph of neighbors centered at a given node within a given radius.

$$S_{u,v} = \begin{array}{c} \begin{array}{c} \text{payment locktime} \\ \text{payment fee ratio} \\ \text{geographic distance} \\ \text{carbon intensity} \\ \text{inter-country hops} \\ \text{inter-continental hops} \end{array} \end{array} \left| \begin{array}{cccccc} x_{u,0,0} & x_{u,1,0} & \dots & x_{u,4,0} & x_{u,5,0} \\ x_{u,0,1} & x_{u,1,1} & \dots & x_{u,4,1} & x_{u,5,1} \\ & & \dots & & \\ x_{u,0,k-1} & x_{u,1,k-1} & \dots & x_{u,4,k-1} & x_{u,5,k-1} \end{array} \right| \# \left| \begin{array}{cccccc} x_{v,0,0} & x_{v,1,0} & \dots & x_{v,4,0} & x_{v,5,0} \\ x_{v,0,1} & x_{v,1,1} & \dots & x_{v,4,1} & x_{v,5,1} \\ & & \dots & & \\ x_{v,0,k-1} & x_{v,1,k-1} & \dots & x_{v,4,k-1} & x_{v,5,k-1} \end{array} \right|$$

Fig. 3. The PCN state representation.

average carbon intensity of the geographic locations of the involved nodes, as defined in Valko and Kudenko (2024).

In the event of a successful payment within a transaction, the agent receives a positive reward, the value of which is dependent upon the selected payment path length and the carbon intensity metric. By optimizing this reward, the agent is able to discern the optimal value of the heuristic modifier, which enables the pathfinding algorithm to identify and dynamically prioritize more efficient payment paths.

The use of average carbon intensity as one of the reward components targets the RL-agent to choose payment paths that minimize the corresponding total carbon intensity and excessive number of inter-continental and inter-country hops or at least maintain a balance between acceptable options. This allows payments to be not only successful, but also effective from the sustainability perspective, and the location based average carbon intensity has already been proven in Valko and Kudenko (2024) as a good metric for this purpose.

The proposed solution is novel in that the agent's actions modify the pathfinding heuristics only within a given range using a discovered latent payment failure distribution. This differs from our preliminary work (Valko and Kudenko, 2023), in which the agent replaced the native pathfinding algorithm. As a result, the payment pathfinding remains transparent and compliant with the approved PCN protocols. From a technical standpoint, this solution preserves scalability while maintaining the same level of privacy of the native pathfinding algorithm. The agent has access only to the dynamic state of a given number of neighboring nodes that are publicly accessible at any given time.

Such pathfinding problems are expected to be effectively addressed by Proximal Policy Optimization (PPO), which is a family of policy optimization RL-based methods that employs multiple epochs of stochastic gradient ascent to perform each policy update. These methods exhibit the stability and reliability of trust-region methods, yet demonstrate superior overall performance and broader applicability (for further details, see Schulman et al., 2017). Furthermore, there is evidence that PPO-based agents are capable of solving specific pathfinding tasks and of adapting such goal-based heuristics more effectively (Skrynnik et al., 2022; Asgari et al., 2022).

The proposed solution employs the *Stable-Baselines3* library, their PPO implementation (Ppo, 2021) and the goal-based learning environment framework (Brockman et al., 2016), extended according to the aforementioned definitions. The hyperparameters recommended for

training in the original PPO paper (Schulman et al., 2017) were used, as they are expected to be effective in most cases (see Appendix A).

3.2. Data and preparation

In this study, LN data snapshots (Decker, 2020) were employed, which have been extensively documented in Zabka et al. (2022). Data collection is based on multiple c-lightning nodes that synchronize their view of the network topology by exchanging *gossip* messages. Each node filters out duplicated or outdated messages and updates its internal view accordingly. To preserve this view across restarts, the node saves these messages in a snapshot file and occasionally compresses it by removing outdated messages (Decker, 2020). Snapshots of the available data can thus be used to reconstruct the network topology at a specific point in time. This approach has been widely employed to study the topology and dynamics of PCNs; see, for example, Zabka et al. (2022), Pietrzak et al. (2021) and Lin et al. (2020).

Based on the aforementioned data and utilizing the *NetworkX* python library, a real network topology was reconstructed and averaged around the most recent available timestamps. Nodes lacking established or deprecated payment channels were excluded, as such nodes are unable to participate in transactions. Therefore, the final graph comprises 59,890 payment channels and 5966 nodes, with a maximum of 1544 neighboring nodes per node.

To ensure a fair comparison with baseline approaches, the same data preparation pipeline as that described in Valko and Kudenko (2024) was reused. The sole distinction pertained to the assignment of a randomly generated failure probability to each payment channel within the reconstructed network, based on the topology and geographic locations. This was done to model the dynamics of a real payment network, with average payment failure rates ranging from 1% to 6%, thus ensuring that the failure probability of inter-continental channels is twice that of inter-country payment channels. The selected range is consistent with pertinent statistics regarding LN failures (see Fig. 4), as well as the statistics of delays in the global network infrastructure resulting from the routing of TCP packets through inter-continental internet channels (Azure, 2024; Chaudhry and Yanikomeroglu, 2022). Therefore, a global distribution of payment failures dependent on the network topology and geographical features was obtained, which is in close alignment with the real world.

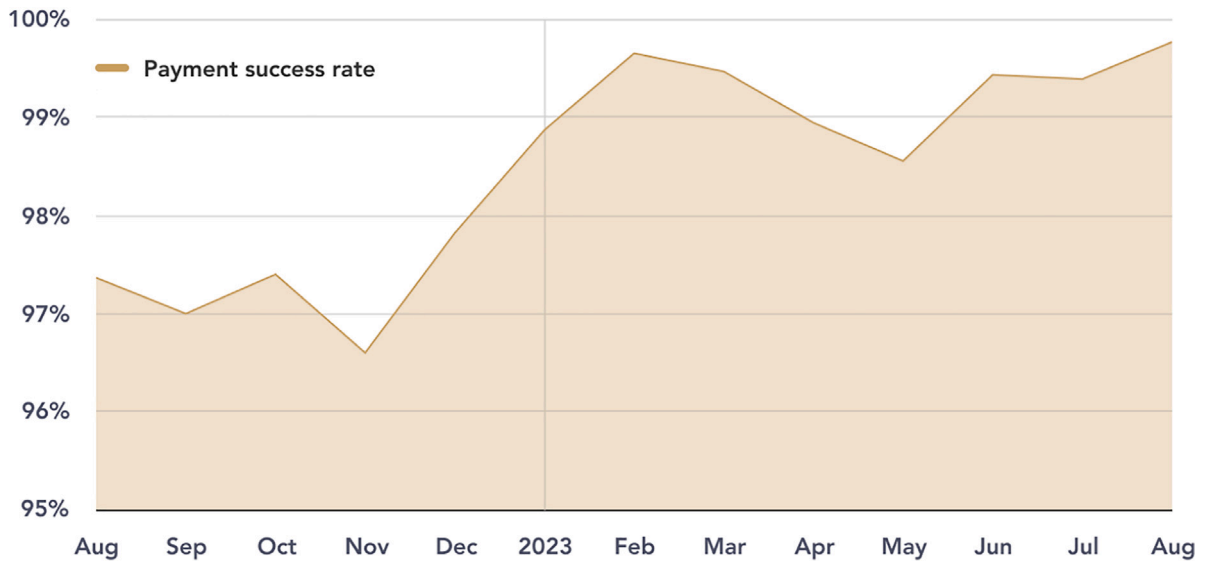


Fig. 4. Payment success rates in LN, %. Note. The most recent statistics regarding the payment success rate utilizing River's LN infrastructure, see [River \(2023\)](#).

In order to assess the efficacy of the proposed approach, 10,000 transactions were pre-generated in a uniform distribution over the network graph, with a logarithmic distribution of payment amounts. These were then divided into a training and test set, with a ratio of 0.7 and 0.3, respectively. The training set was then employed to train a RL-agent utilizing each of the three native algorithms, namely LND, CLN and ECL. As the algorithms in question employ disparate pathfinding heuristics (for further details, see [Kumble and Roos, 2021](#); [Kumble et al., 2021](#)), three distinct types of agents were developed, each trained in an environment utilizing one of the native algorithms. A comparison of the performance of these agents was deemed necessary in order to select the most efficient one, given that it is evident that the policies learned by them are disparate.

3.3. Evaluation strategy and compared algorithms

As the review has demonstrated, the extant deterministic algorithms (e.g., Flash) exhibit inferior performance with respect to payment success relative to the actual statistics of the LN with the native algorithms (see [River, 2023](#)). Additionally, known RL-based solutions are not designed to address the payment pathfinding problem directly and cannot be meaningfully compared with native algorithms without substantial revisions or retraining. It was thus resolved to assess the efficacy of the proposed solution in conjunction with the native algorithms, as well as with their enhanced version following static optimization, which serves as a baseline.

As previously stated, there are three principal LN software clients with integrated solutions based on Dijkstra's algorithm: LND, CLN and ECL. These have an open-source python implementation ([Kumble and Roos, 2021](#); [Kumble et al., 2021](#)) that has been early adapted to run with modified heuristics ([Valko and Kudenko, 2024](#)).

In the preliminary work, it was demonstrated that the topology and geographical distribution are significant factors for the LN. Furthermore, the pathfinding heuristics of the native algorithms can be adaptively modified to achieve sustainability goals. For this purpose, the modified heuristic and the specific optimal values were proposed.³ In addition to the native LND-, CLN- and ECL-heuristics, this modified

heuristic h introduces a minor cost to the native cost functions in the event that the payment nodes are situated in disparate countries or continents. Consequently, the corresponding carbon intensity is increased. The value of h , which provides a gentle incentive for pathfinding algorithms to perform location-based searches ([Valko and Kudenko, 2024](#)), is employed as a baseline approach for this work with Eq. (2).

$$h(u, v) = \eta(u, v) \cdot \frac{int_{CO_2}[v] + int_{CO_2}[u]}{2} + (1 - \eta(u, v)) \cdot (int_{CO_2}[v] - int_{CO_2}[u]), \quad (2)$$

where $[u, v]$ is a payment channel (or path as a sequence of such channels) that is established from node u to node v ; int_{CO_2} is the average carbon intensity of involved nodes' geographic locations; η is the heuristic modifier. The modifier value is typically set in the range of $[-1..1]$, as defined in [Valko and Kudenko \(2024\)](#).

Notwithstanding the favorable outcome of the static optimization process, which met the requisite sustainability criteria, there has been no testing of the results under dynamic infrastructural payment failure conditions. Furthermore, static optimization techniques are typically computationally expensive, rendering them unsuitable for deployment in a solution that encompasses every payment transaction.

The experimental evaluation strategy entails the execution of native algorithms, namely *native_{LND}*, *native_{CLN}*, *native_{ECL}*, and the algorithms with enhanced heuristics (h) after static optimization, namely *baseline_{LND}*, *baseline_{CLN}*, *baseline_{ECL}*. Since the proposed solution utilizing dynamic RL-agent-based optimization (denoted as *improved_{LND}*, *improved_{CLN}*, *improved_{ECL}*), it is evaluated on a fixed pre-generated set of transactions, employing a snapshot of a real network with a hidden distribution of payment failures (see Section 3.2). In order to obtain robust estimates, all of the examined algorithms are to be run five times on the same test set of transactions, but with different random seeds due to the presence of stochastic components. Subsequently, the payment success rates, defined as the number of successful payments divided by the total number of payments, are averaged across all runs and algorithms. These rates are then analyzed against the pre-defined probability of payment failure using statistical criteria, fitted regressions and visual analysis. The two-sided Mann-Whitney U-test with standard levels of significance was used for statistically valid comparison of average values because it is based on ranks and the median and does not require normality for the data distribution.

³ The special heuristic that minimizes the average carbon intensity for the selected payment channels yielded the following results: $\eta_{LND} = 0.27876$, $\eta_{CLN} = 0.25556$, $\eta_{ECL} = 0.35784$ ([Valko and Kudenko, 2024](#)).

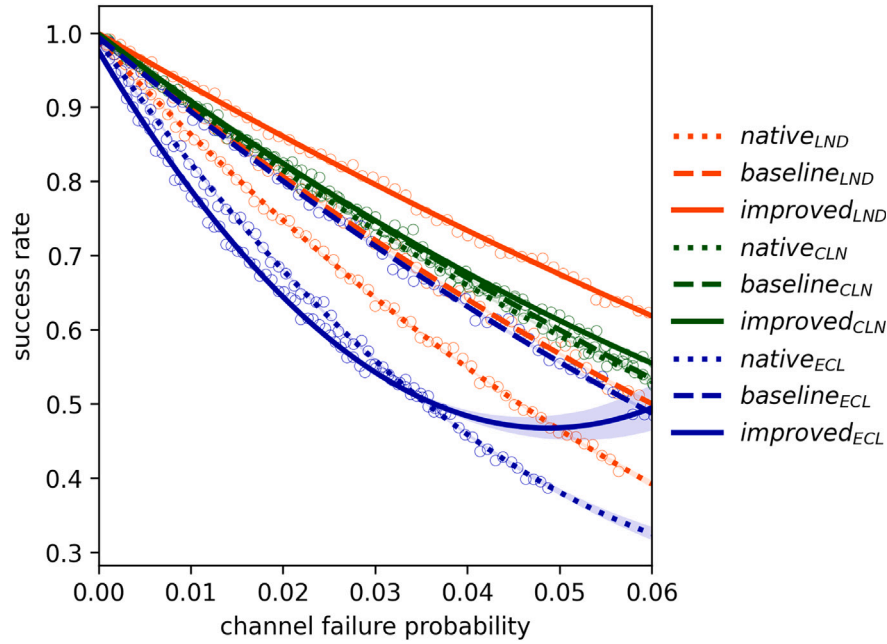


Fig. 5. Payment success rates comparison.

Note. The figure presents the payment success rates for three approaches: the *native* algorithms, the *baseline* solution based on static optimization, and RL-agents trained on the corresponding native algorithms (*improved*). The experiments were repeated five times using the same transaction test set but with different random seeds, leading to different channel failure probability distributions. The shaded area beneath the fitted curves represents the 95% confidence intervals.

Table 2
Payment success rates comparison using statistical criteria.

	$native_{LND}$	$baseline_{LND}$	$improved_{LND}$	$native_{CLN}$	$baseline_{CLN}$	$improved_{CLN}$	$native_{ECL}$	$baseline_{ECL}$	$improved_{ECL}$
$native_{LND}$	0.660								
$baseline_{LND}$	–	0.673							
$improved_{LND}$	***	**	0.760						
$native_{CLN}$	*	–	–	0.724					
$baseline_{CLN}$	*	–	–	–	0.718				
$improved_{CLN}$	**	*	–	–	–	0.753			
$native_{ECL}$	–	–	***	**	**	***	0.628		
$baseline_{ECL}$	–	–	**	–	–	*	–	0.667	
$improved_{ECL}$	–	–	***	*	–	**	–	–	0.654

Note. The figure highlights the statistical significance of differences in the median payment success rates among the *native* algorithms, the *baseline* solution after static optimization, and *RL – agents* trained on the corresponding native algorithms (*improved*). The experiments were repeated five times using the same transaction test set but with different random seeds, resulting in different payment channel failure probability distributions.

* The Mann–Whitney two-sided *U*-test was used, significance levels: $p < 0.05$.

** The Mann–Whitney two-sided *U*-test was used, significance levels: $p < 0.01$.

*** The Mann–Whitney two-sided *U*-test was used, significance levels: $p < 0.001$.

4. Results

4.1. Payment success rates comparison

As previously stated, the hypothesis was that a well-trained agent could achieve superior results to those obtained through static multi-criteria optimization, particularly in the presence of hidden dynamic failure probabilities. To test this hypothesis, a test sample of transactions was used to compare the baseline approach with the results achieved by each trained RL-agent in terms of payment success rate. To corroborate the dynamic stability of the proposed solution compared to native algorithms and the baseline approach, the average probability of payment failure was varied within a range of 1% to 6% and randomly distributed across the network (see *Methods*).

As illustrated in Fig. 5 and Table 2, the results demonstrate that among the *native* algorithms, CLN exhibits the highest reliability, with a median payment success rate of 0.724 in the selected range of the probability of payment failure. This is followed by LND and ECL, with median success rates of 0.660 and 0.628, respectively. The *baseline*

approach (after static optimization) demonstrates superior estimates, albeit by a marginal degree. The results for baselines of CLN, LND, and ECL are 0.718, 0.673, and 0.667, respectively.

These results demonstrate notable discrepancies between the existing pathfinding heuristics of the native algorithms and simultaneously substantiate the efficacy of static pathfinding optimization. However, further optimization is possible, as the mean payment success rate decreases markedly as the probability of payment channel failures rises (Fig. 5).

The proposed *improved* solutions demonstrate superior performance, yet RL-agents trained on disparate native algorithms exhibit markedly disparate outcomes. The ECL-based RL-agent ($native_{ECL}$) demonstrated the poorest performance, with a payment success rate of 0.654. In contrast, the CLN-based agent ($improved_{CLN}$) exhibited a relatively strong performance, with a payment success rate of 0.753 (see Table 2). The LND-based agent ($improved_{LND}$), however, demonstrated the best performance. The median success rate within the selected failure probability range is 0.760, which represents a 5% improvement over the best median performance of the native algorithms.

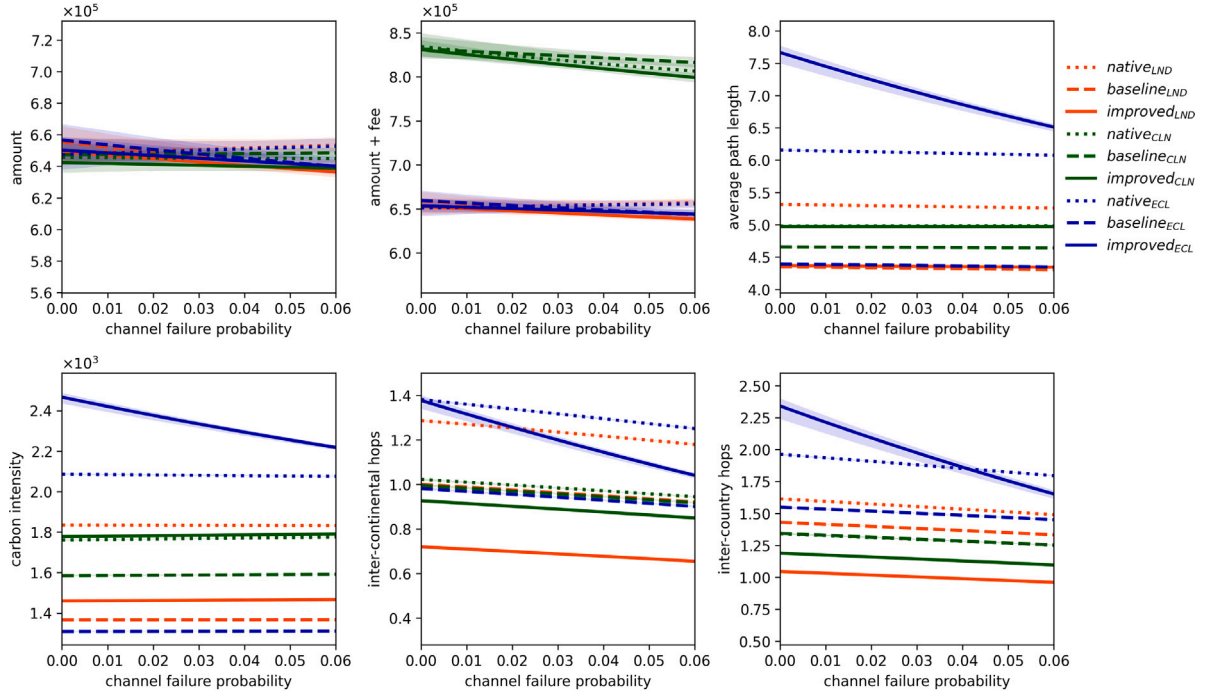


Fig. 6. Total network throughput and average payment path length comparison.

Note. The figure presents the total network throughput and average payment path length for three approaches: the *native* algorithms, the *baseline* solution based on static optimization, and RL-agents trained on the corresponding native algorithms (*improved*). The experiments were repeated five times using the same transaction test set but with different random seeds, resulting in different payment channel failure probability distributions. The shaded area beneath the fitted curves represents the 95% confidence intervals.

It should be noted that the median values provide a simplified overview of the data, whereas a closer examination of the regression curves may offer more insight into the actual differences. The approximation of the empirical data demonstrates a notable disparity in performance among the analyzed solutions in the context of increasing payment failure probabilities, see Fig. 5. In particular, the agent trained on LND, with a payment failure probability of approximately 1% per payment channel, achieves a success rate of 97%, which is very close to the baseline solution. However, as the failure rate increases, the payment success rate of the proposed solution decreases at a slower rate than that of the competitors. With an average failure rate of more than 5%, it outperforms the other solutions by more than 10%, which represents a highly competitive result.

Table 2 presents a similar comparison, but based on the Mann-Whitney statistical criterion. In the context of this two-sided version, the null hypothesis is that there is no difference. Overall, while the applicability of this criterion can be subject to scrutiny, the outcomes align with the visual analysis. The RL-agent trained on the LND algorithm (*improved_{LND}*) demonstrates the highest payment success rate or performs not worse than the other agents in all comparisons. Although it performs better numerically in all cases, there is a minor statistical difference from the CLN-trained agent that becomes significant when using the one-sided version of the criterion. This can be attributed to the superior performance of the LND algorithm itself in comparison to other native algorithms (for further details, see Kumble and Roos, 2021), thus additional dynamic optimization enhances its efficacy.

Therefore, the proposed approach for dynamic optimization of pathfinding heuristics demonstrates satisfactory reliability results in the context of an increasing average payment failure probability. Note that a comparison of the results obtained with other algorithms, such as Flash, and with other known RL-based solutions is not performed for reasons discussed in the evaluation strategy section (see Section 3.3). These reasons relate to the fact that state-of-the-art algorithms already

perform worse than native LN algorithms in a comparable setup (see, e.g., Wang et al., 2024; Chen et al., 2022a).

4.2. Multi-criteria comparison

In practice, increasing payment success probabilities should be considered as a goal together with minimizing fees. Naturally, these goals can be contradictory, since node operators can and will choose fees freely. Unfortunately, achieving both of these goals are weakly NP-hard problems (Pickhardt and Richter, 2021).

The objective of multi-criteria optimization is to achieve a compromise between the general characteristics of payment transactions, such as total payment amounts and fee ratios, while simultaneously enhancing more crucial target metrics, including payment path length, the number of inter-continental and inter-country hops, and average carbon intensity. To this end, a detailed comparison of these metrics was conducted for the proposed solution (Fig. 6, see also Appendix B).

The agent was trained using a reward function that employs only the most pertinent metrics, including payment success, the length of the selected payment path, and the carbon intensity. It would be inaccurate to describe this case as a full multi-criteria optimization, given that other criteria, such as the payment fee ratio, also have significance from the user's perspective. Nevertheless, preliminary research has demonstrated that although the utilization of multiple multi-directional target metrics in the design of an agent's reward is theoretically feasible, in practice it impairs the agent's ability to learn effectively.

In consideration of the most significant user metric – payment fee ratio and the most significant economic metric – total payment throughput, it can be concluded that in all cases, RL-agents demonstrate performance that is not significantly inferior to that of the baseline approach (Fig. 6). Furthermore, with regard to sustainability metrics (e.g., payment path length, number of inter-continental and inter-country hops), they even exhibit superior performance, as evidenced

in Fig. 6, see also Appendix B. It is anticipated that metrics of low importance may be sacrificed in order to achieve a higher payment success rate. For example, *improved_{CLN}* has the second-best average payment success rate but ends up using payment paths with higher fees (although the *improved* version is still better than *baseline*). Generally, the RL-agent trained on the LND algorithm (*improved_{LND}*) also performs better in this comparison, as it achieves higher payment success rates and shorter path lengths without significantly reducing overall throughput.

Therefore, although the objective of this paper was not to pursue a full multi-criteria optimization, the proposed solution exhibits favorable comparison results and can be regarded as a compelling proof of concept.

4.3. Scalability and sustainability considerations

Simulation experiments show that the proposed approach performs effectively on a snapshot of a real full-size network, yielding results comparable to native algorithms currently used in client software, indicating that it is appropriately scalable with respect to network size.

To provide an approximate assessment of the scalability of the proposed hybrid architecture in terms of execution time, the average runtime per transaction and per payment hop, including the pathfinding phase, was compared (see Fig. 7).

Fig. 7 demonstrates that the computational complexity of python implementations for algorithms within the hybrid architecture increases only slightly (it takes less than a second on average to execute a single transaction). This is primarily due to the relative simplicity of policy-based inference within the trained deep RL-model. Typically, for a neural network with L layers and P parameters per layer, the computational complexity is roughly $O(L \cdot P)$ for a forward pass. Consequently, the approximate performance in runtime also indicates an acceptable technical scalability of the proposed solution. Further improvement in RL-model inference latency is possible with the model compression approach, in some cases it can reach more than 50% (McEnroe et al., 2024; Valko and Kudenko, 2023), but this is beyond the scope of this work.

It is difficult to estimate the real carbon footprint for globally distributed networks that dynamically utilize different client hardware. According to one source, execution of one LN transaction with 6 payment hops results in emission of about 0.00992 gCO₂eq. Barratt and Scott (2021), and thus has a significant negative impact on the environment. Of course, the emission dynamics in such a network depends on the network traffic, the number of transactions at a time, and the performance of the native pathfinding algorithms.

To give a raw estimate of the environmental impact of the proposed improvements, the transmission of relevant TCP network messages through the worldwide network infrastructure can also be considered. First, there is a direct correlation between quality of service (QoS) in a TCP communication network and carbon emissions. Specifically, carbon emissions increase as QoS (Habib and Marimuthu, 2013) deteriorates. This growth is attributed to the increase in the number of packets generated and their subsequent transmission over an unstable network. A study of a modestly sized network of 100 nodes operating under varying loads and maintaining a QoS of at least 80% showed an average emission rate of 0.156 gCO₂eq. per byte (Usman et al., 2017). This figure is an annual average based on the use of natural gas as the primary source of electricity. However, it is difficult to get a more accurate estimate due to the use of different network equipment (Usman et al., 2017). According to reference data, the peak power consumption of modern Ethernet switches, which mainly perform the packet routing function, is approximately $4.42E-09$ gCO₂eq. per byte (He, 2019; Vishwanath et al., 2015).

Second, the maximum packet length size in the LN is 65569 bytes.⁴ Thus, having estimates for the average infrastructure use and worst case packet size, assuming for simplicity that one hop requires one maximum length packet, executing one transaction of 5 hops results in 51.14 kgCO₂eq.

Given that the proposed solution reduces the average path length by ~ 3 hops, including ~ 1 inter-country/inter-continental hop, the result is a $\sim 60\%$ reduction in the message transmission footprint or 30.69 kgCO₂eq. per average transaction. In addition, using average carbon intensity as one of the reward components and a key optimization metric minimizes the excess emissions associated with the number of inter-continental/inter-country hops and uneven infrastructure usage, making the potential for carbon footprint reduction apparent.

5. Limitations and reproducibility

The present experimental setup inherits some of the limitations of the data used and the methods of their processing, thereby ensuring comparability with the previously obtained results.

The first limitation of this study and its experimental design is the dynamic and sparse nature of the LN, which may result in slight discrepancies in the numerical results obtained from different snapshots. A second limitation is inherent to the geolocation method, which relies on public IP addresses. A significant number of LN nodes utilize onion and hidden addresses. Nevertheless, the visible sub-network is sufficiently large to allow for an analysis of the network's topological features, as previously demonstrated by Howell et al. (2023) and Valko and Kudenko (2024). A further issue that requires consideration is the impact of modifying native pathfinding algorithms on security and privacy of the system (Cai et al., 2020), thus further research is required to investigate potential privacy and security issues in light of known vulnerabilities in the LN, as identified in the previous studies (Beres et al., 2020; Kappos et al., 2021). However, it is important to note that this is an emerging research area (Malavolta et al., 2017; Kappos et al., 2021; Kumble et al., 2021). The chosen baseline approach (see Valko and Kudenko, 2024), which prioritizes localized payment channels, is aligned with the idea (Kumble et al., 2021) that sensitive information should not traverse unknown payment channels that are either poorly described, lack public metadata, or are significantly remote.

Deep reinforcement learning, which we employ in this study, has made notable advancements across a range of tasks. However, security and privacy concerns associated with deploying robust RL applications remain a topic of ongoing debate (for recent discussions, see Mo et al., 2024). For instance, an adversary may manipulate the environment to influence an agent's actions, thereby misleading it to behave in an anomalous manner.

At this stage, we utilized a RL-agent to adapt native pathfinding heuristics, whose computation algorithms are already documented in open source. This approach is intended to prevent malicious code injection (for further details, see Kumble et al., 2021; Kumble and Roos, 2021). Nevertheless, despite the transparency of the code, the native pathfinding algorithms exhibit some degree of ambiguity, thereby providing a technical safeguard against the potential reconstruction of payment paths. In this context, the RL-agent introduces an additional element of uncertainty with regard to the extent of exploitation of the solution space, which is fixed and can be configured by the agent's parameters. It is therefore our contention that the deployment of RL-based methodologies does not result in a diminution of the aforementioned security measures compared to the baseline solution. However, this is a topic that warrants further investigation and discourse.

⁴ <https://github.com/lightning/bolts/blob/master/08-transport.md#lightning-message-specification>.

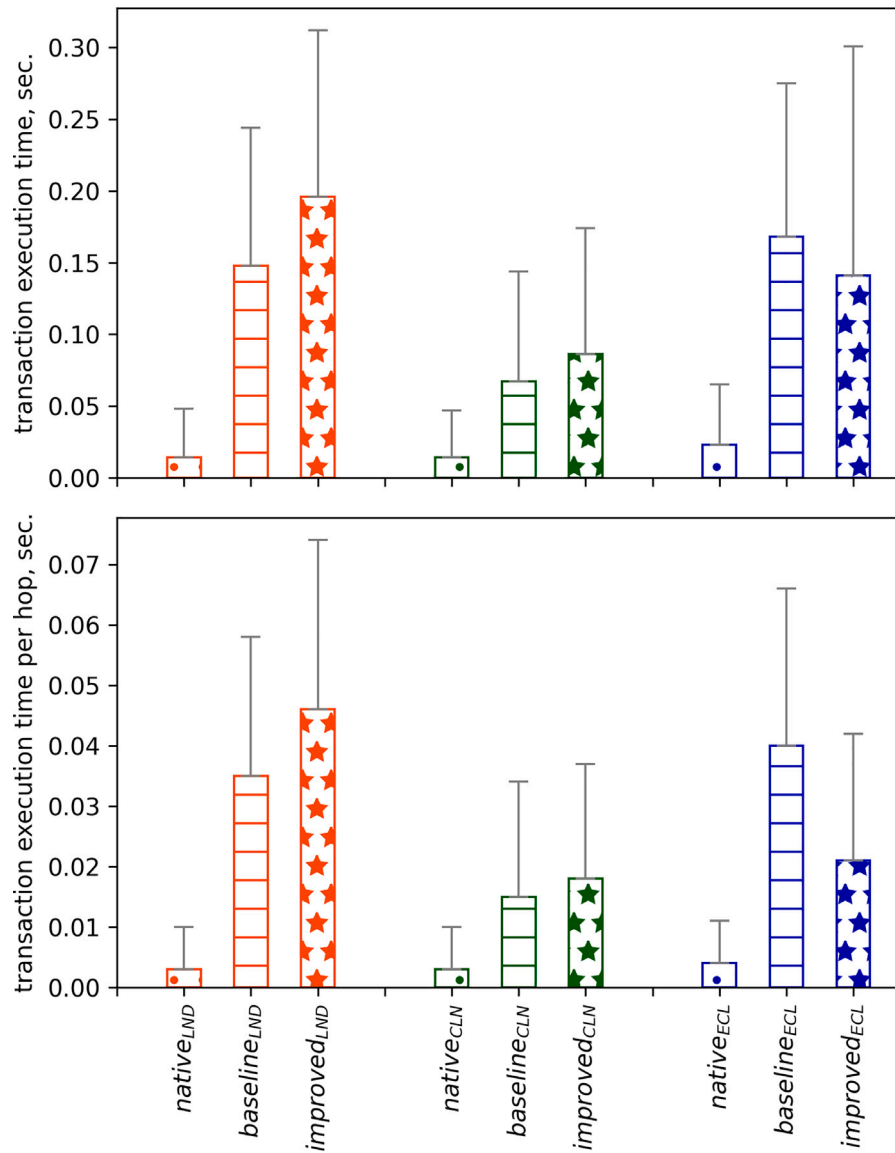


Fig. 7. Transaction execution time comparison.

Note. The figure shows the average execution time of a successful payment transaction, including the pathfinding phase, for three approaches: the *native* algorithms, the *baseline* solution, and RL-agents trained on the corresponding native algorithms (*improved*). The experiments were repeated five times using the same test set of 3000 uniformly distributed transactions but with different random seeds.

6. Conclusion and outlook

The application of reinforcement learning to solve payment pathfinding problems in different types of networks has been the subject of extensive study in the literature. Nevertheless, its deployment in globally distributed peer-to-peer PCNs, such as the LN, has predominantly concentrated on profitable solutions, while the issue of payment success has remained relatively unresolved, especially in the context of multi-criteria optimization and balancing carbon neutrality of infrastructure. The objective of this paper is to evaluate, contrast and enhance the existing pathfinding algorithms in the LN through the utilization of reinforcement learning, with the aim of optimizing payment success and, consequently, enhancing infrastructure sustainability. This work makes a contribution to the field by considering reinforcement learning as a dynamic heuristic improvement approach to achieve robust payment efficiency in a geographically distributed network infrastructure.

Through the lens of applying deep reinforcement learning to the pathfinding task in peer-to-peer PCNs, this paper introduces a novel hybrid architecture that combines a reinforcement learning agent with native deterministic pathfinding algorithms. This hybrid approach is scalable enough and outperforms native algorithms and state-of-the-art static optimization methods, delivering enhanced reliability and efficiency.

The subsequent stages of this research will concentrate on enhancing multi-criteria dynamic optimization and incorporating it into existing LN software clients, with comprehensive real-world testing, as well as further investigating of security and privacy concerns. It would be beneficial for future work in this area to focus on further developing environmentally and topologically sustainable algorithms and heuristics with the aim of achieving robust, carbon-neutral payment infrastructure.

Table 3

Multi-criteria comparison numerical results.

Target metrics	<i>native</i> _{LND}	<i>baseline</i> _{LND}	<i>improved</i> _{LND}	<i>native</i> _{CLN}	<i>baseline</i> _{CLN}	<i>improved</i> _{CLN}	<i>native</i> _{ECL}	<i>baseline</i> _{ECL}	<i>improved</i> _{ECL}
<i>Path length</i>	5.316 ± 1.321	4.345 ± 0.871	4.363 ± 0.859	4.983 ± 1.149	4.654 ± 1.039	4.975 ± 1.193	6.152 ± 1.506	4.394 ± 0.932	7.963 ± 7.008
<i>Geographic distance</i>	10 941.629 ± 7870.366	8460.456 ± 6886.841	6688.297 ± 6162.966	8732.885 ± 6873.177	8455.904 ± 6777.604	8380.934 ± 6763.812	12 527.681 ± 8726.082	8463.207 ± 6955.616	14 895.842 ± 19 991.512
<i>Carbon intensity</i>	1836.134 ± 554.085	1366.109 ± 397.369	1458.250 ± 418.626	1762.521 ± 518.335	1581.665 ± 476.663	1779.814 ± 546.738	2083.946 ± 602.682	1308.847 ± 378.119	2525.357 ± 1680.744
<i>Payment locktime</i>	191.246 ± 96.911	211.327 ± 111.678	209.613 ± 115.670	124.620 ± 60.742	131.781 ± 63.985	130.787 ± 62.862	250.477 ± 127.799	218.043 ± 114.684	397.016 ± 389.451
<i>Payment fee ratio</i>	1.014 ± 0.266	1.018 ± 0.267	1.018 ± 0.269	1.225 ± 0.803	1.134 ± 0.638	1.129 ± 0.621	1.014 ± 0.266	1.018 ± 0.267	1.026 ± 0.270
<i>Number of inter-continental hops</i>	1.296 ± 1.034	1.006 ± 0.857	0.722 ± 0.749	1.028 ± 0.899	0.998 ± 0.880	0.929 ± 0.857	1.388 ± 1.075	0.986 ± 0.869	1.445 ± 1.787
<i>Number of inter-country hops</i>	1.623 ± 1.166	1.434 ± 1.003	1.047 ± 0.939	1.347 ± 1.014	1.348 ± 1.023	1.190 ± 0.990	1.971 ± 1.372	1.556 ± 1.057	2.502 ± 3.829

Note. The figure shows the mean values and standard deviations ($\pm$) of the target metrics for three approaches: the *native* algorithms, the *baseline* solution based on static optimization, and RL-agents trained using the native algorithms (*improved*). The experiments were repeated five times using the same transaction test set but with different random seeds, leading to different channel failure probability distributions in each run.

CRedit authorship contribution statement

Danila Valko: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Data curation, Conceptualization. **Daniel Kudenko:** Writing – review & editing, Writing – original draft, Validation, Supervision, Conceptualization.

Code and data availability

All data and the developed algorithms can be accessed via the public repository: <https://github.com/ellariel/ln-dynamic-optimization>.

Using generative AI and AI-assisted technologies

During the preparation of this work the authors did not use any generative AI and AI-assisted technologies in the writing process.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Experiment setup details and hyperparameters

The following hyperparameters for the *Stable-Baselines3* PPO implementation (Ppo, 2021) were used: policy type: actor-critic; timesteps per epoch: 50,000 for general training; learning rate: 0.000001; discount factor (γ): 0.99; GAE parameter (λ): 0.95; clipping parameter: 0.2; value function coefficient: 0.5; maximum value for gradient clipping: 0.5; 4 vectorized environments; random seed is fixed.

Hardware setup for training. CPU: AMD EPYC 7662 64-Core Processor, 256 CPUs; GPU: 2 x A100-PCIE-80 GB; 1 TB RAM, 1007.764 GB available; Platform system: Linux-5.10.0-15-amd64-x86_64-with-glibc2.31; Python version: 3.10.6.

Hardware setup for experiments. CPU: 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40 GHz, 8 CPUs; GPU: 1 x NVIDIA GeForce MX350; 8 GB RAM, 7.675 GB available; Platform system: Windows-10.0.22621-SP0; Python version: 3.9.13.

Appendix B. Multi-criteria comparison details.

See Table 3.

Data availability

All data and the developed algorithms can be found in the public repository.

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